



Lab 3: Kirchhoff's Current Law (KCL), Current Divider Rule (CDR) using Parallel and Ladder Circuit

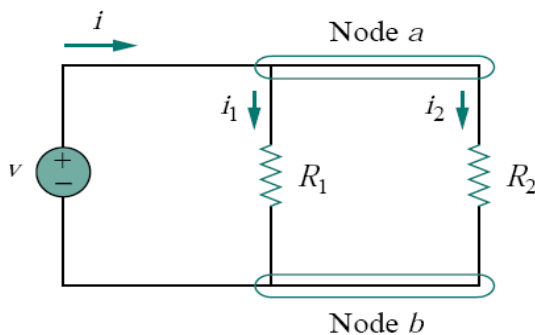
Objectives

- Learn how to connect a parallel circuit on a breadboard.
- Validate the current divider rules.
- Verify Kirchhoff's current law.
- Verify KCL and KVL in ladder circuit.

Theory:

Current Division Rule: The total current i is shared by the resistors in inverse proportion to their resistances. This is known as the principle of current division, and the circuit in Figure. 1 is known as a current divider.

For two resistors in parallel:

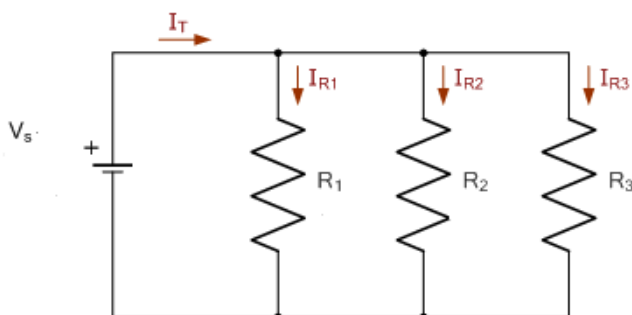


In this circuit:

$$i_1 = \frac{R_2}{R_1 + R_2} i$$

$$i_2 = \frac{R_1}{R_1 + R_2} i$$

For three resistors in parallel:



In this circuit:

$$R_{eq} = \left[\frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \right]$$

$$I_T = \frac{V_s}{R_{eq}}$$

$$I_{R1} = \frac{R_{eq}}{R_1} \times I_T$$

$$I_{R2} = \frac{R_{eq}}{R_2} \times I_T$$

$$I_{R3} = \frac{R_{eq}}{R_3} \times I_T$$

Kirchhoff's Current Law: Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.

Mathematically, KCL implies that
$$\sum_{n=1}^N i_n = 0$$
 Where, N is the number of branches connected to the node and i_n is the n^{th} current entering (or leaving) the node.

An alternative form of KCL: The sum of the currents entering a node is equal to the sum of the currents leaving the node.

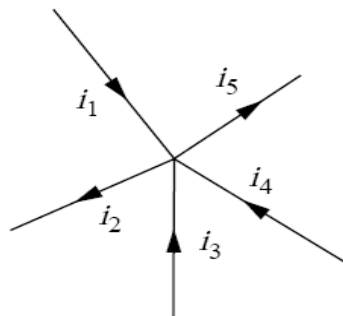


Figure.1 Currents at a node illustrating KCL

From the above figure we see that, currents i_1 , i_3 , and i_4 are entering the node, while currents i_2 and i_5 are leaving it. By applying KCL we get,

$$i_1 - i_2 + i_3 + i_4 - i_5 = 0$$

Rearranging terms gives,

$$i_1 + i_3 + i_4 = i_2 + i_5$$

Ladder Circuit: The ladder circuit represents a commonly used circuit style that is configured purely on the basis of series and parallel connections. It is named as because it resembles a ladder.

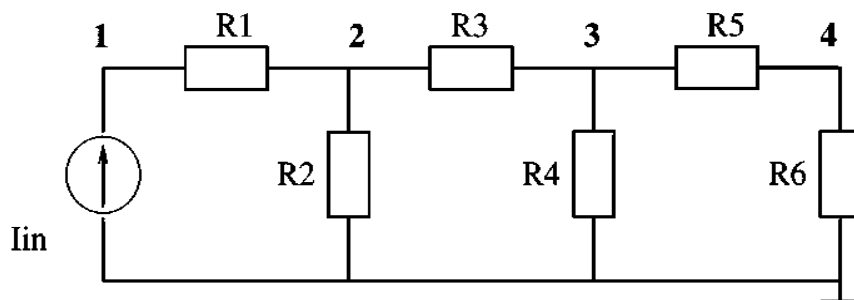
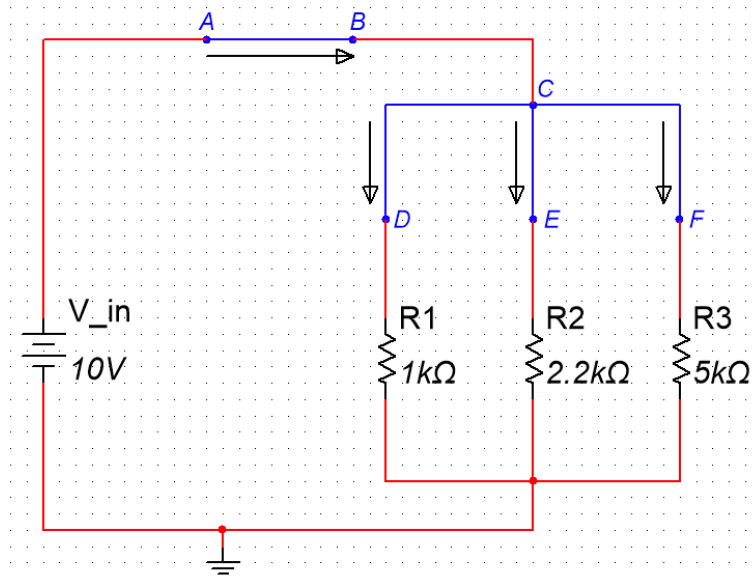


Figure.4 A three section ladder circuit

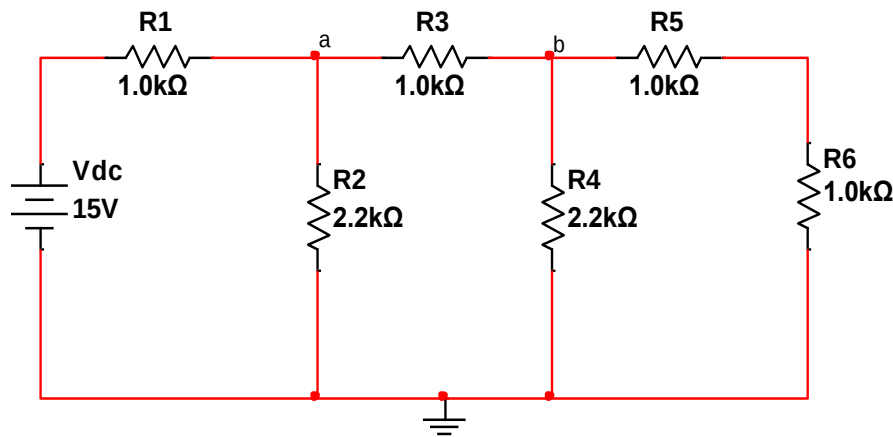
List of Components:

Component Name	Value (or Model Number)	Quantity
Breadboard		1
DC Power Supply (Variable)		1
Resistors	1.0 kΩ	1
	2.2 kΩ	1
	5.0 kΩ	1
Connecting Wires		10
Digital Multimeter (DMM)		1

Circuit Diagram:



Circuit 1



Circuit 2

Procedure:

1. Identify all the given resistors using color coding and fill in the required columns in Table 1.
2. Measure the resistances of the resistors using the DMM and fill in the required column in Table 1.
3. Calculate the percentage error of the resistance values.

$$\text{Percentage of Error} = \frac{|(\text{Practical value} - \text{Theoretical value})|}{\text{Theoretical value}} \times 100\%$$

4. Build the circuit 1
5. Connect the DMM (current mode) in series,
 - between A and B to measure I_{in}
 - between C and D to measure I_1
 - between C and E to measure I_2
 - between C and F to measure I_3
6. Record the readings in Table 2. Then fill in Table 3.
7. Construct Circuit 2.
8. Using a DMM, measure the potential differences across all the resistors in circuit 2. Record all the readings in Table 4
9. Using a DMM, measure the current through all the resistors and record in Table 4.

Data Collection:

Lab 3:

Group No. _____

Instructor's Signature _____

Table 1:

Theoretical Value	Practical Value using DMM	% of Error
1 k Ω		
2.2 k Ω		
5 k Ω		
$R_T = R_1 \parallel R_2 \parallel R_3 = 604.3 \Omega$		

Table 2:

	Experimental readings	Theoretical values	% of Error
I_{in}			
I_{R1}			
I_{R2}			
I_{R3}			

Table 3: use data from table 2

Supply Current I_{in}		Is Total Current equal to sum of branch current?
Sum of Branch Current $I_{R1} + I_{R2} + I_{R3}$		☐ Equal ☐ Almost Equal ☐ Not Equal

Table 4: For Circuit (2)

Component	Voltage	Current
V_{dc}		
R1		
R2		
R3		
R4		
R5		
R6		

Questions:

1. State the current division rule (CDR).
2. State the Kirchhoff's current law (KCL).
3. With the experimental data, verify Kirchhoff's voltage law within each independent closed loop of the circuit.
4. Showing all steps, calculate the theoretical values in Table 2. Compare theoretical values to your experimental values and explain whether your circuit follows KCL or not.
5. Showing all the steps, theoretically calculate R_{eq} . Compare with the experimental value.
6. Calculate all the theoretical values for Circuit 2 - Table 4. Show all steps.

MultiSim Simulation

- Simulate the KCL circuits you did in lab class. Fill the data tables. Verify KCL.
- Design the following ladder circuit on MultiSim. Then find the voltages across each resistor, current through each resistor using Multimeter. Then calculate the power absorbed by each resistor. Show data on a table.

